

Rohan Mehra

Graduate Student in Computer Science

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Education

- 2025 – present **MS in Computer Science, ISIMA, Clermont Auvergne INP, UCA**
- Ranked: 8/34
 - *Selected Courses:* Advanced Machine Learning and Data Mining, High Performance Computing, Decision and Learning under Uncertainty.
- 2021 – 2025 **BS in Electrical Engineering and Computer Science, IISER Bhopal**
- Grade: 8.92/10
 - *Selected Courses:* Data Science and Machine Learning, Multi-agent Reinforcement Learning, Algorithms, Operating System, Computer Organization, Theory of Computation, Probability and Statistics, Linear Algebra, Competitive Programming.

Research Experience

- Mar. – present **LIFAT, CNRS, Université de Tours, France, M2 Research Intern**
- 2026
- Advised by Prof. Gilles Venturini as part of RFAI team.
 - Working on AI-assisted image recognition of bumblebees (*bombus*).
 - Using transfer learning, and explainable AI approaches.
- May 2025 – **Institut Pascal, CNRS, Université Clermont Auvergne, France, Research Intern**
- Mar. 2026
- Advised by Dr. Mathieu Labussière as part of ComSee team.
- & May – July 2024
- Worked on deep learning models on SWIR images' for road perception in adverse weather conditions.
 - Models worked on include Grounded SAM, YOLO, RF-DETR [Sem. Segmentation] and CUT, CycleGAN, Pix2Pix, BBDM [GANs & diffusion-based]
- Jan. – April 2025 **IISER Bhopal, India, BS Project Student**
- Advised by Dr. Arpit Sharma at the Dept. of EECS.
 - Worked on using Zero-Shot Learning model for Sector Classification, utilizing the CrowdRE dataset.
 - Used models including Sentence-BERT, All-MiniLM-L12, Bert4RE and BERTOverflow
- May – July 2023 **Indian Institute of Space Science and Technology, Research Intern**
- Advised by Prof. Deepak Mishra as part of the CVVR Lab
 - Worked on robotics and implementation of SLAM through Kalman filter and EKF Methods.
 - Used Robot Operating System (ROS), RViz, and Gazebo.

Selected Publications

Full list available on [Google Scholar](#)

- 🔗 Rohan Mehra, Alexandre Riffard, Yannis Loumouamou, Mathieu Labussière, "Analysis of SWIR Imaging Detection Performance Under Adverse Environmental Conditions for Autonomous Driving Systems", **Under review at IEEE Transactions on Intelligent Vehicles (T-IV)**, 2026.
- 🔗 Yannis Loumouamou, Alexandre Riffard, Rohan Mehra, Mathieu Labussière, "Experimental study of YOLOv8 fine-tuning for RGB-SWIR multimodal road detection", *Reconnaissance des Formes, Image, Apprentissage et Perception (RFIAP)*, 2026.
- 🔗 Rohan Mehra, Alexandre Riffard, Mathieu Labussière, Pierre Duthon, Romuald Aufrère, "Would SWIR modality help for detection and segmentation in harsh weather conditions? An experimental study.", *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2025. [Oral]

Teaching Experience

- 2024 & 2023 **Basic Electronics, IISER Bhopal** ($\times 2$)
- Teaching Assistant for an undergraduate class of 180 students for 2024-25/23-24 Semester I.
 - Conducted tutorial sessions, prepared assignment solutions, and provided one-to-one guidance.

Academic / Course Projects

- 2025 **Risk-Aware Multi-Epoch Package Delivery Planning**, IISER Bhopal
- 🔗 ○ Used greedy algorithm to solve package delivery problems while balancing risk and reward.
 - Developed a GUI-based visualization tool using Tkinter and Matplotlib.
- 2024 **Training Multi-Robot Warehouse (RWARE) Environment**, IISER Bhopal
- 🔗 ○ Trained robots for optimized coordination in the RWARE environment.
 - Employed Deep Q-Network (DQN) for efficient learning.
- 2023 **Prediction of Mortality Rate of Heart Failure Patients in ICU**, IISER Bhopal
- 🔗 ○ Built a classification model to predict ICU mortality in heart failure patients.
 - Utilized Random Forest, Logistic Regression, and AdaBoost etc. for model development.

Technical Skills

Languages Python (with PyTorch framework), C++, C, Java
Software ROS, Gazebo, MATLAB, Git, $\text{\LaTeX}$, Wolfram Mathematica

Events / Fellowships / Achievements

- 2026 **ELLIS Winter School on Foundation Models (FoMo)**
Selected to attend the FoMo Winter School hosted by the ELLIS unit Amsterdam, among a global pool of applicants (~ 250 participants).
- 2025 **MIAI Cluster Excellence Scholarship**
Selected for Excellence Scholarship program offered by the Multidisciplinary Institute in Artificial Intelligence (MIAI) in France to pursue master's program, awarded to ~ 20 candidates university wide.
- 2024 **IASc-INSa-NASi Summer Research Fellowship** [Declined]
Selected for summer research fellowship jointly offered by the Indian Academy of Sciences, Indian National Science Academy, and National Academy of Sciences, awarded to the top $\sim 5\%$ of applicants.
- 2021 **Indian Maritime University Undergraduate Admissions**
Secured All India Rank 9 out of 14,108 ranked candidates, placing in the top 0.1% nationwide.

External / Online Courses

- CS231n: Deep Learning for Computer Vision - (Offered by Stanford Online)
- Introduction to Deep Learning & Neural Networks with Keras – Coursera (Offered by IBM)
- Image Processing Onramp, Simulink Onramp, MATLAB Onramp – MathWorks

Positions of Responsibility / Extra Curriculars

- 2023 – 2024 **Cultural Secretary, IISER Bhopal**
Elected representative of the student body for organizing all cultural events at IISER Bhopal. Led a team of 70 performers to the the Inter IISER Cultural Meet 2023 held at IISER Mohali.
- 2022 **NCC 'B' Certificate, National Cadet Corps, India**
Completed 'B' division NCC military training from 8 UP BN NCC, Aligarh.

Other

Languages English (fluent, TOEFL 99/120), Hindi (Native), French (Basic)
Hobbies Table Tennis, Quizzing